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VOLCANOLOGY

A clearer view of the current phase of unrest at Campi Flegrei caldera

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We used a retrained machine learning workflow to enhance the performance of the seismic monitoring network at Campi Flegrei caldera (CFc) for improved tracking of the evolution of volcanic unrest. We analyzed the recent (21 January 2022 to 20 March 2025) continuous seismic data, which showed a sharp increase in seismicity at the highly populated CFc. Our analysis expanded the seismicity catalog from around 12,000 to more than 54,000 earthquakes. The more complete picture of seismicity revealed a sharply defined caldera ring fault system (RFS) featuring a very narrow depth range of seismicity, clearly illuminated shallow faults in the northern part of the caldera, and uncovered a source of very shallow hybrid earthquakes likely related to the hydrothermal system. To date, we have not observed any direct signature of upward migration of magma, nor have we observed seismic activity at depths below 3.7 kilometers on or inside the RFS.

Campi Flegrei caldera (CFc) is an active volcanic field that poses a direct threat to several hundred thousand people residing within the caldera and an additional million people in the nearby city of Naples, which could be severely affected by heavy ash fall and pyroclastic flows in the case of an eruption. The most recent CFc eruption, in 1538, was preceded by decades of unrest characterized by the uplift of the caldera floor, seismic swarms, and variations in fumarolic activity (*1*, *2*). Centuries of quiescence were interrupted by recent activity, which includes four episodes of unrest in the last 75 years: 1950–1952, 1969–1972, 1982–1984, and 2005–present (Fig. 1B). These episodes were characterized by increased seismicity and uplift of the central part of the caldera, near the Pozzuoli port. The most recent phase began in 2005, with a cumulative maximum uplift of about 1.35 m (October 2024), and a sharp rise in seismicity since August 2023, resulting in thousands of recorded earthquakes per month, including five of duration magnitude (M_d) ≥ 4.0 (Fig. 1). This latest phase of unrest has raised concerns among scientists, the public, and decision-makers regarding the potential reactivation of the volcanic system as well as the direct seismic threat posed by shallow, locally damaging earthquakes. The processes driving both the seismicity and uplift remain a topic of debate characterized by conflicting interpretations of monitoring data (*3*–*6*). We applied a machine learning workflow (*7*, *8*) to provide a more comprehensive and clearer view of the space-time evolution of recent seismicity at CFc, revealing important aspects of the earthquake generation process and its potential link to magmatic activity.

The current phase of unrest

The ongoing unrest at CFc has persisted for two decades, exhibiting considerable fluctuations but showing increasing rates of seismic activity, uplift, and fumarole emissions. Unlike the most recent Bradyseismic crisis of 1982–1984, during which the Pozzuoli port experienced uplift of 1.8 m (*9*) in less than 2 years, the current episode has seen slower but sustained uplift of 1.4 m over 20 years (Fig. 1B). This period has also been marked by a major increase in the areal extent and gas flux from the Pisciarelli fumarole (*10*). Other key indicators, such as the seismicity distribution, geochemical composition, and temperature of fumaroles, as well as the bell-shaped pattern of ground deformation (*11*), show no clear differences from the previous unrest, although subtle changes could be difficult to discern owing to changes in monitoring. During the 1982–1984 crisis, ground deformation was tracked through leveling campaigns, which recorded the same bell-shaped deformation pattern observed today but with less-frequent sampling (*12*, *13*). Regarding geochemical parameters, only the CO_2/H_2O and CH_4/H_2O ratios and H_2S have been consistently monitored at the Solfatara fumarole since May 1983. These ratios show trends similar to those observed during the current unrest, particularly since 2020, when CH_4/H_2O levels began to increase considerably (*11*, *14*). However, for the 1982–1984 unrest, the observed H_2S excursion was modest compared with that measured since 2018 (*15*). After steady decadal uplift, seismicity increased dramatically in mid-2023 (Fig. 1C). By the end of 2022, the absolute uplift exceeded levels reached in the previous phase of unrest (*16*). In addition to possible future volcanic activity, seismic activity has raised concerns about seismic risk. The earthquakes have been moderate in size; however, some have produced remarkably intense ground shaking close to the epicenter because of their shallow depth (*17*).

Operated by Istituto Nazionale di Geofisica e Vulcanologia - Osservatorio Vesuviano (INGV-OV), the seismic network increased in the number of stations and areal extent between January and July 1984, from 4 in 1982 to more than 15 in mid-1984 (*18*). The number decreased after 1984 and increased again in 2000, with a large increment since 2018, to reach the current number of 38. These new deployments are predominantly broadband, with some being very broadband as well (*5*).

After the increase in seismicity in mid-2023, the nonseismic elements of the monitoring network were greatly expanded (*5*) to include 35 global navigation satellite systems (GNSS) stations distributed across the caldera, including offshore areas; 10 tiltmeters; five tide gauges; two strainmeters; two fixed thermal cameras; and three multiflux geochemical stations observing the Solfatara and Pisciarelli fumaroles. Regular gravimetric, thermal, and geochemical campaigns have further enhanced the monitoring (*11*).

The driving forces behind the current phase of unrest are under debate. Recent studies suggest that these phenomena are driven by magmatic injection from a deep reservoir into a magma chamber at ~4 km depth (*3*, *19*). These findings are based on modeling ground deformation data, which examines the behavior of a continuous and spatially heterogeneous elastic volume influenced by potential sources at varying depths. However, these models do not explain why CFc seismicity lacks the seismic signatures that are expected to accompany magmatic injection from a deep magma chamber at 8 km depth (*20*).

Improving the earthquake catalog

Machine learning methods have enabled dramatic improvements in the effectiveness of seismic monitoring (*7*, *21*), including the seismicity of magmatic systems (*22*, *23*). In this study, the PhaseNet machine learning model was retrained on CFc earthquake and noise waveforms and then applied (see supplementary materials) to continuous seismic data from the monitoring networks of Campi Flegrei (fig. S1) from 21 January 2022 to 20 March 2025. We used 48 seismic stations located in the caldera, with 30 stations deployed during the analyzed period (fig. S2). The Gaussian Mixture Model Association (GaMMA) method uses soft clustering to associate observed arrival times with the earthquake sources that generated them. Improved locations are developed in a two-step process (as described in the supplementary materials). The improved workflow captures the spatiotemporal evolution of seismic activity more completely and can detect

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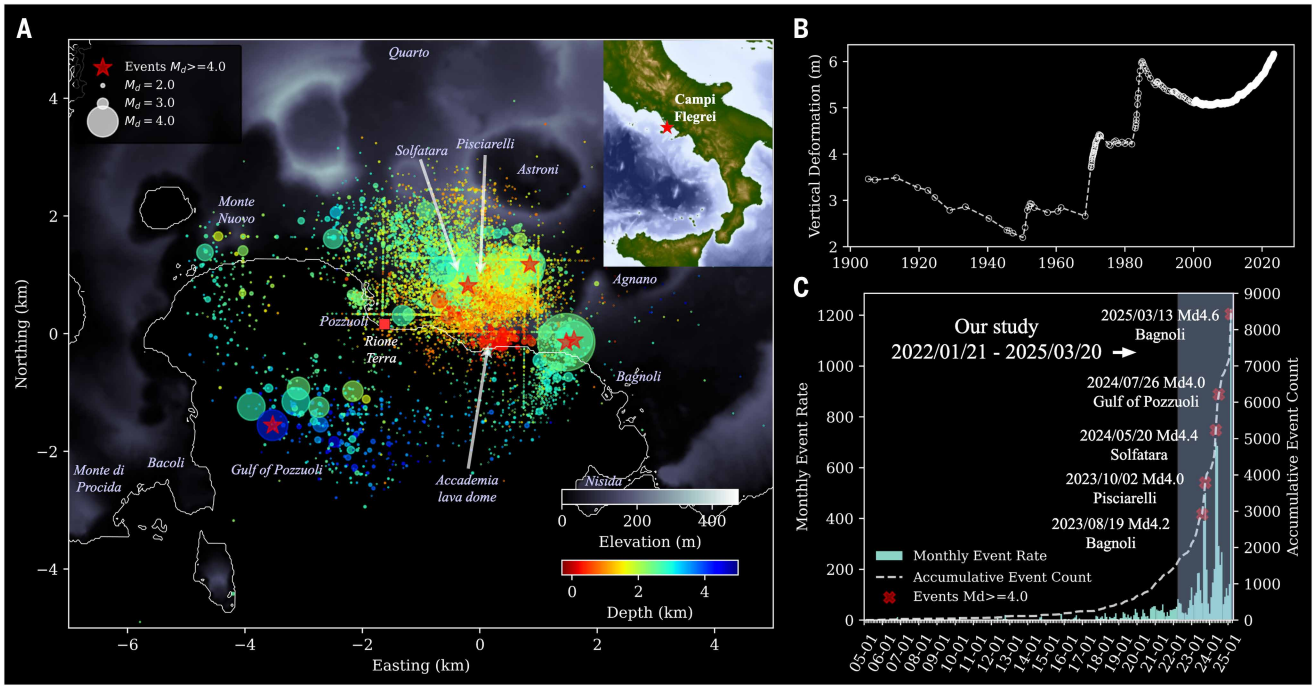


Fig. 1. General seismicity and deformation of Campi Flegrei caldera. (**A**) Map of Campi Flegrei caldera (CFc) showing the seismic events (circles) from 1 January 2005 to 20 March 2025 included in the Istituto Nazionale di Geofisica e Vulcanologia - Osservatorio Vesuviano (INGV-OV) catalog (*24*). Red stars mark five events with duration magnitude (M_d) ≥ 4.0. Coastlines and the inland 0-elevation contour (sea level) are indicated by the solid white line. Coordinates are relative to the reference point (14.14°E, 40.82°N), and depths are measured relative to sea level. (**B**) Vertical ground displacement recorded from 1905 to 2023 at Pozzuoli port, Rione Terra [red square in (A)] (*11*,*52*,*53*). (**C**) Seismicity evolution of CFc from 2005 to 2025. The study period (21 January 2022 to 20 March 2025) is shaded in white. The cumulative seismicity curve only includes events larger than completeness magnitude 0.2 (M_d) (*5*).

possible magmatic signatures that can provide crucial insights into the evolving state of the caldera. We have operationalized these methods so that improved information is available at the observatory on a daily basis.

Over the past 3 years, we detected and located 54,319 earthquakes (Fig. 2), which is four times as many as the 12,083 manually located by the seismological laboratory of INGV-OV (*24*) (fig. S5). Our phase-picking and association methods in particular enhanced the detection of low-magnitude and overlapping events. Furthermore, by using the double difference approach (*25*), with cross-correlation-based arrival times (*26*), and a three-dimensional (3D) velocity model (*27*), we improved earthquake location precision to reveal a highly resolved view of CFc seismicity (Figs. 2 and 3). In particular, the ring fault is sharply resolved around most of its circumference, extending over a very limited depth range and featuring several notable gaps in activity. The catalog also resolves multiple subparallel faults onshore under Pozzuoli and shows that the depth extent of seismicity across the entire caldera is limited to <3.7 km.

A clearer view of the current phase of unrest

Overall seismicity at Campi Flegrei during this unrest is increasing, with more frequent swarms and bigger events (Fig. 2) (*28*). In early 2022, most seismic events were concentrated onshore at shallow depths near Solfatara and Pisciarelli, with few events occurring on the ring faults (fig. S10A). In the more recent 2023–2025 swarms, different sectors of the ring system were activated (Fig. 2B and figs. S10 and S11).

Seismicity of the ring fault system (RFS hereafter) was clearly defined by our catalog as having a very narrow depth range, locally no more than several hundred meters from top to bottom and deepening relative to sea level from 2.1 to 3.7 km from north to south (Fig. 3). The RFS includes a prominent straight segment on its eastern boundary and two gaps—one on the south and one on the northeast. It follows offshore bathymetric lines (*29*) and the distribution of the Secca delle Fumose fumaroles in front of Monte Nuovo. This submarine bathymetric feature delineates the fault system that hosted the initial stages of the 1536 eruption (*30*). The RFS also bounds the northern part of the Solfatara-Pisciarelli hydrothermal system.

From a dynamic point of view, the RFS closely fits the geometry of a hypothesized resurgent block (*9*) and gravimetric Bouguer anomaly (*31*). The new catalog helps delineate the south-sloping boundary of a planar structure within the central zone of the caldera. Focal mechanisms are consistent with the dynamics of a resurgent block centered in the caldera, bounded by vertical faults on the east side and high-angle reverse faults on the southwest side (Fig. 4A). Recent seismic slip has likely amounted to a small fraction of the observed caldera uplift during this time period.

As regards the depth distribution of RFS earthquakes, deep exploratory drill holes at San Vito and Mofete (~4 km to the north and to the west of the caldera center, respectively) (*32*) and multiple geophysical surveys (*6*, *32*, *33*) identify major differences in the structure and physical state of the caldera in the depth range of the ring. Temperatures in the deepest drill holes reach 350°C at a depth of 3000 m (*32*). Below ~3000 m, the stratigraphy also changes from a caldera fill consisting of intermixed sediments and tuffites, tuffs, and lavas to thermally metamorphosed precaldera carbonates (*34*). Compression wave–to–shear wave (Vp/Vs) ratios <1.6 at these depths revealed by tomographic studies carried out at different times since the early 2000s (*20*, *34*–*36*) imply low bulk modulus relative to a Poisson solid, likely caused by high gas pressure in the ring's interior that drives the shallow seismicity (*34*). The limited depth range of ring seismicity seems to be stable throughout the study period, without marked vertical or horizontal migration

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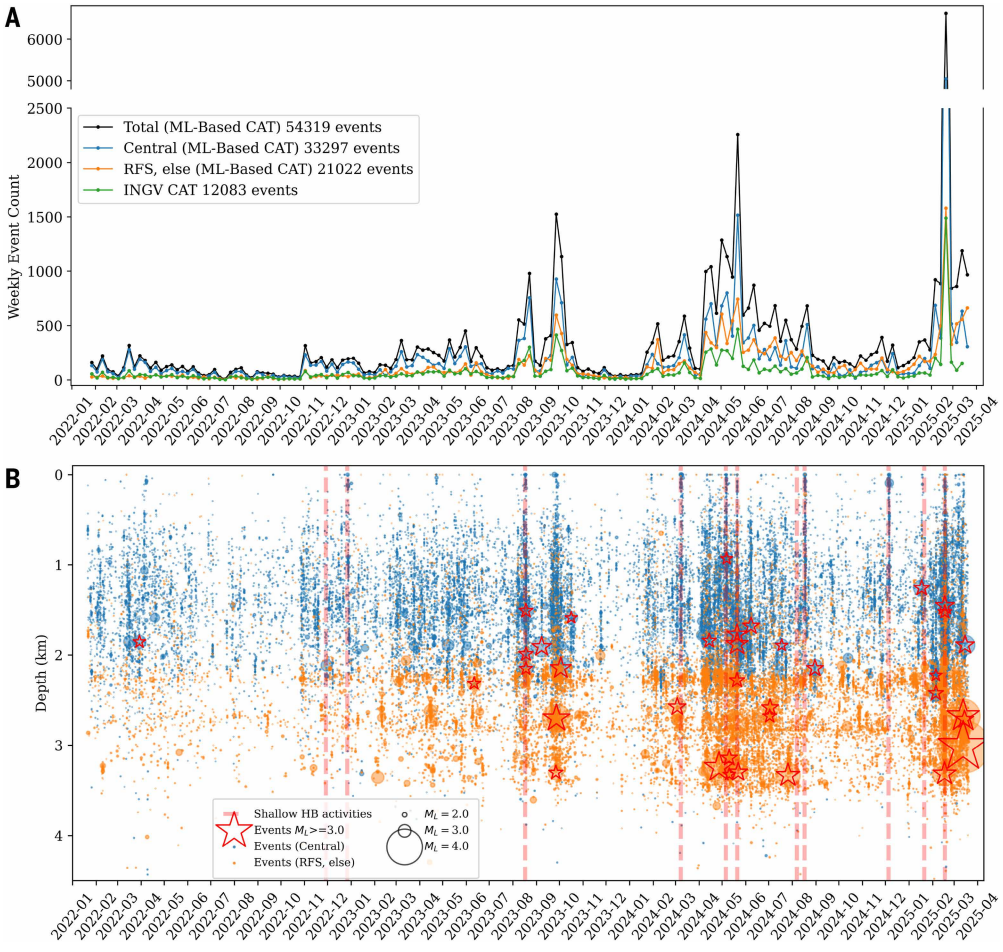


Fig. 2. Overview of the Campi Flegrei seismicity. (**A**) Seismicity rate computed with the relocated machine learning (ML)–based catalog (CAT) and INGV-OV routine catalog (*24*). Curves show the seismicity rate of the central region (blue) and RFS plus other regions (orange). These two regions are separated by the dashed borders in Fig. 3. The *y* axis includes a break from 2500 to 5000 to show the largest swarms in 2025 without compressing the rest of the series. Detection was particularly improved during major swarms. (**B**) Relocated ML-based earthquake depth distribution over time, with event magnitude denoted by the size of the dot. Horizontal lineations result from persistent depth-limited seismicity across different sectors (also seen in Fig. 3). The 0-km mark indicates sea level. Near-surface seismicity (0 to 0.5 km) is concentrated in the Accademia lava dome, and shallow seismicity (0.5 to 2.1 km) mostly occurs in the Pozzuoli-Solfatara-Pisciarelli-Agnano region. Seismicity is also observed at depths of 2.1 to 2.5 km in the northern sector of the RFS, 2.5 to 3.0 km in the eastern sector, and 2.5 to 3.7 km in the southwestern sector. Events reported below 3.7 km are uncertain and likely located this deep because of inaccurately measured arrival times (see supplementary materials). Shallow hybrid (HB) events (figs. S14 and S15) are denoted with dashed red lines. M_L, local magnitude.

(Fig. 2B and fig. S10). This apparent stable behavior can be explained by different mechanisms. For example, a brittle-ductile transition caused by rapidly increasing temperature with depth could define the lower limit of seismicity (*37*), whereas the liquid-gas phase transition may mark its upper limit (*38*), which is also sharply bounded. An alternative explanation is that at these temperatures, the deeper metacarbonate layer may be in the brittle field (*39*); in this case, gas overpressure at 3 to 4 km depth may be locally destabilizing the RFS in the precaldera basement.

In the ring's interior, shallow seismicity mainly locates beneath the hydrothermal area of Solfatara and Pisciarelli. This activity is more dispersed than on the RFS; however, earthquake locations reveal a series of subparallel east-northeast–to–west-southwest trending faults reaching the Pozzuoli port. This fault orientation is confirmed by the focal mechanisms calculated for that area (Fig. 4A). Shallow faults beneath the Solfatara crater and the Pisciarelli fumarole align with zones of elevated CO_2 flux (*40*), indicating that these structures likely serve as pathways for ascending gas and hydrothermal fluids. A similar system was observed during the 1989 unrest at Mammoth Mountain, California (*41*). The shallowest earthquakes locate at <1 km depth and include hybrid events (Fig. 2B and figs. S14 and S15) under the Accademia and Monte Olibano lava domes. We interpret these as the consequence of microcracking phenomena that involve the domes on the southern part of the Solfatara-Pisciarelli hydrothermal system. Fracturing processes allow the passage of hydrothermal fluids to resonate within the opened cracks (*42*), producing the hybrid seismogram character of these events. Giudicepietro *et al.* (*43*) identified a geodetic anomaly in the behavior of the Accademia and Monte Olibano lava domes starting in 2021, deviating from the bell-shaped uplift characteristic of Campi Flegrei bradyseism. This anomaly has been interpreted as evidence of a mechanical discontinuity within the medium induced by stress associated with the ongoing inflationary phase (*43*).

Implications for the physical process driving the unrest

Despite the intensifying unrest, we do not observe clear magmatic signatures or evidence of large-scale space-time migration in the recorded seismicity. Specifically, we observe only very restricted hybrid and long-period earthquake activity, concentrated at very shallow (<1 km) depths under the Accademia and Monte Olibano domes (figs. S14 and S15) and clearly related to the hydrothermal system (Fig. 2). Although the central onshore seismicity expands over a broad area over time (Fig. 2 and fig. S10), we do not observe migration across caldera sectors. Similarly, although the RFS seismicity exhibits intensified activity from 2022 to 2025, it remains spatially localized (Fig. 2 and fig. S10). The same fault segments are repeatedly reactivated during different swarms, illuminating consistent structural patterns (fig. S11). Indeed, similarities with the spatial pattern of seismicity during the 1982–1984 episode (*9*, *44*) suggest a similar physical mechanism that has produced earthquakes on the same structures.

Upward fluid migration in magmatic systems is thought to be accompanied by seismic signals (*45*, *46*), and Giacomuzzi *et al.* (*47*) have suggested that CFc seismicity at depths of 3.5 to 4 km is a signature for inflation of a shallow magmatic body. However, common-receiver gathers of earthquakes from different RFS sectors recorded at CPOZ, a station located at the caldera center (fig. S12), show clear *S*-wave phases in all the traces, supporting the absence of magmatic bodies at depths shallower than 3.5 to 4 km. We note the discovery of a large magmatic body at a depth of 8 km and the reported absence of melt bodies >1 km^3 above this depth (*20*, *34*), as determined through analysis

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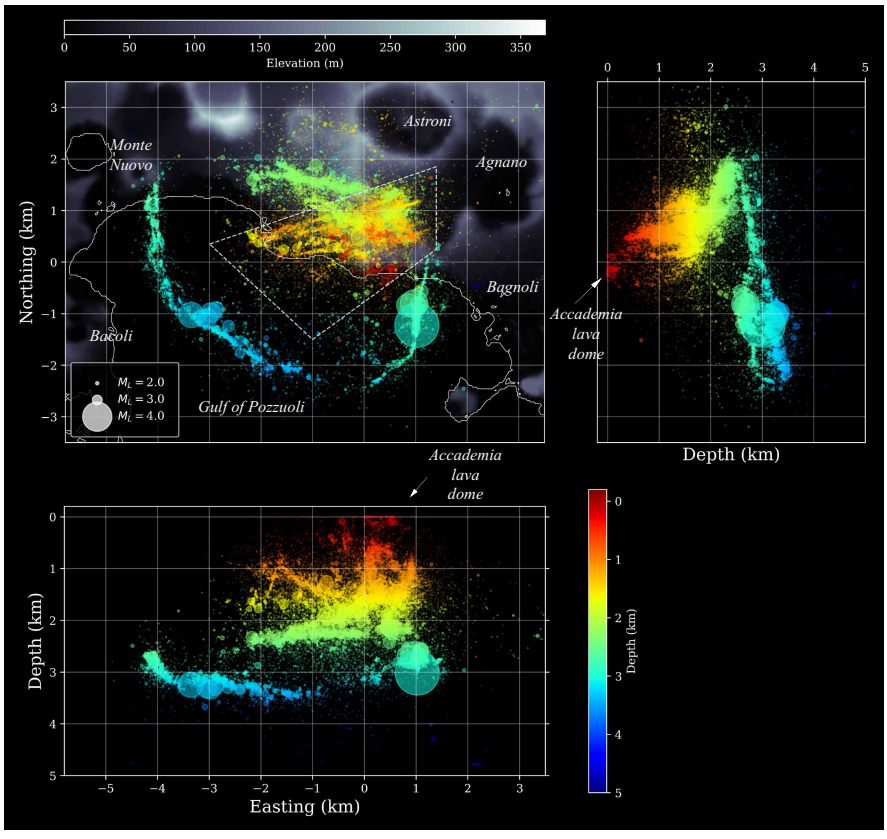


Fig. 3. Spatial distribution of seismicity at Campi Flegrei caldera. The top left panel shows the map view, the top right panel shows the depth versus northing cross section, and the bottom panel shows the depth versus easting cross section. In the map view panel, the region inside the dashed border is referred to as the central zone in Fig. 2. Coastlines and the inland 0-elevation contour (sea level) are indicated by the solid white line. Coordinates are relative to the reference point (14.14°E, 40.82°N), and depths are measured relative to sea level.

New perspectives for improved seismic risk assessment

In addition to the volcanological implications, the enhanced high-resolution earthquake catalog can inform seismic risk assessment, which is thought to be the most immediate threat, as demonstrated by the request made by the mayor of Naples to a task force at the University of Naples (*17*) to carry out an expeditious seismic risk assessment to inform mitigation strategies.

A reliable seismic risk assessment depends on understanding both the magnitude-frequency distribution and the spatial distribution of sources capable of generating damaging earthquakes. Magnitude-frequency distribution describes the statistical distribution of earthquake magnitude, which is an essential ingredient to estimate the probability of potentially damaging events. Meanwhile, the spatial distribution is typically informed by a synthesis of mapped faults identified in the scientific literature and an earthquake catalog.

The high-resolution machine learning–based earthquake catalog presented in this paper offers considerable advancements in both aspects of seismic risk assessment. During the current earthquake sequence, the operational procedures at the INGV-OV have provided duration magnitudes for the earthquakes identified through classical procedures. The approach adopted for the new high-resolution earthquake catalog yields direct estimates of local magnitudes for a markedly higher number of detected earthquakes. These local magnitudes follow an exponential distribution above the completeness magnitude, which is consistent with expectations for a homogeneous earthquake catalog (fig. S8).

of data collected between 1982 and 2001. The compositional analysis of fumarolic gases within the caldera suggests that they originate primarily from decompression-driven degassing of magma at depths >6 km, accompanied by partial remobilization of sulfur from hydrothermal mineral phases (*15*). Contrary to previous studies (*9*, *47*–*50*), we detect no seismic sources below 3.7 km on or inside the RFS. This lack of activity is not necessarily due to the mechanics of the lower crust because seismic events in magmatic systems are known to occur under increased strain rate and/or increased pore pressure. Although major magma injection up to a depth of 3.7 km without accompanying seismic signals is possible, such an occurrence would be remarkable.

The *b*-value of the magnitude-frequency distribution is similar to that observed in tectonic environments (fig. S8), which is in agreement with the absence of very shallow magmatic bodies that are usually evidenced by the presence of high *b*-values. Moreover, despite the different resolutions, the spatial pattern of seismicity appears similar to that in the 1982–1984 phase of unrest (*9*, *51*), suggesting that most of the seismicity we detect occurs on preexisting faults (*17*).

Figure 2B provides important insights into the potential sources of seismicity. After periods of reduced activity in mid-2022, early summer 2023, late 2023, and late 2024, we observe a relationship between seismic activity at RFS and within the caldera during subsequent increases in activity. Notably, these surges do not include very shallow events (<500 m), which seem unassociated with ring activity. This pattern suggests that seismicity tends to resume after a lull of several months, which may approximate the time required for pressure within the ring to reach critical effective normal stress.

Furthermore, the high-resolution catalog delineates more precisely the location, depth, and geometry of active 3D structures (Fig. 3) that could host damaging events. These can be integrated into fault mapping efforts (*17*) to help refine the dimensions and orientations of CFc faults identified by various methods. Mapping hazardous tectonic structures is particularly crucial for seismic risk assessment, given the characteristics of the current earthquake sequence in Campi Flegrei: Owing to the typically shallow depths of hypocenters, the most severe damage is often observed near the epicenters (*17*). Consequently, the enhanced spatial representation of potential active seismic sources depicted in Figs. 3 and 4 should enable seismologists to prioritize potential event locations by expected damage. Specifically, several of these structures intersect densely populated areas. Notably, multiple kilometer-long, subparallel faults in the central zone extend from the Pozzuoli port toward residential zones west of Monte Spina (fig. S13B). As for the RFS, its northern sector lies beneath the A56 highway, and both the northeastern and northwestern ring sectors reach the Bagnoli and Lucrino urban zones (fig. S13A), respectively. Although many RFS sectors are located offshore, the occurrence of a large event could be quite hazardous, considering their shallow depths and their proximity to inhabited areas. For example, the offshore 13 March 2025 M_d 4.6 event caused damage in the town of Bagnoli. These long, offshore fault segments could potentially host future earthquakes with magnitudes >4, considering the intensifying seismicity observed during the ongoing unrest.

We observe that structures that have hosted $M_d \geq 4.0$ events (Fig. 1) are illuminated by small events early in the 2022–2025 sequence. For

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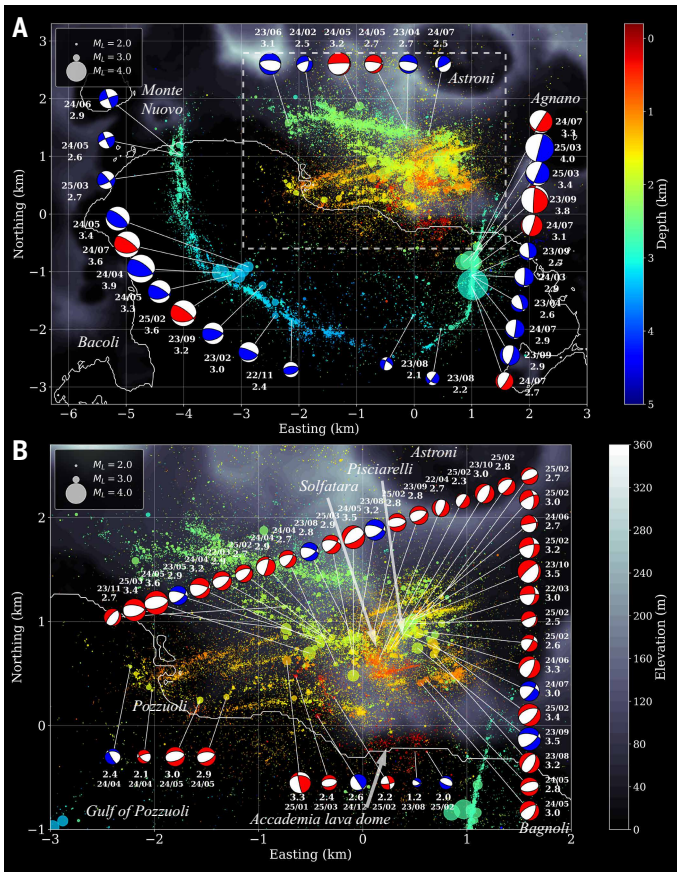


Fig. 4. Focal mechanism solutions of larger events. (**A**) Seismicity and focal mechanism solutions for larger events within the ring-shaped seismicity zone. Each event is labeled with its occurrence date ("YY/MM") and local magnitude; dilatational quadrants are shown in white on the lower hemisphere. Coordinates are relative to the reference point (14.14°E, 40.82°N). (**B**) Zoomed-in view of the seismicity and focal mechanism solutions in the Pozzuoli-Solfatara-Pisciarelli-Agnano region and the Accademia lava dome. Focal mechanisms are calculated using gCAP (*54*–*56*) and SKHASH (*57*). Focal mechanism solutions classified as high quality (see supplementary materials) are plotted in red; all others are plotted in blue.

instance, the central normal faults and the eastern RFS sector were seismically active in 2022 (fig. S10A), before the 19 August 2023 Bagnoli earthquake (M_d = 4.2) and the 12 October 2023 Pisciarelli earthquake (M_d = 4.0). Similarly, seismicity along the southwestern RFS sector was clearly delineated in 2023 (fig. S10B), preceding the 26 July 2024 M_d 4.0 earthquake.

Earthquakes continue to strike the local communities of Naples as of this writing, including one of the strongest to date in the current phase, a M_d 4.4 on 13 May 2025 in the shallow offshore zone beneath Pozzuoli, and a M_d 4.6 on 30 June 2025 on the southwest sector of the offshore RFS (*11*). Both events occurred on structures that had hosted smaller events in our catalog (Fig. 4), highlighting the value of machine learning–enabled catalogs for identifying faults that may yet fail as long as the caldera remains in unrest.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Figs. S1 to S15; Tables S1 and S2; References (*62*–*75*); Data S1 to S3; Interactive Plot S1

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A clearer view of the current phase of unrest at Campi Flegrei caldera

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Editor's summary
The millions of people who live near Campi Flegrei, the "burning fields," in Naples, Italy, likely do not feel most of the tens of thousands of earthquakes that have occurred there occurred since August 2023; however, after centuries of quiet, the caldera seems to be waking up. With data from regional monitoring, Tan *et al*. retrained a machine learning model to define the depths of nearly 50,000 earthquakes that have occurred in the Campi Flegrei caldera since 2022. Their maps bring into sharp focus where seismic activity has accelerated in urban zones along shallow lineaments and around the caldera ring. —Jake S. Yeston

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